

Designing Hybrid Unboxing with Labubu: Configurable Suspense in Digital–Physical Reveals

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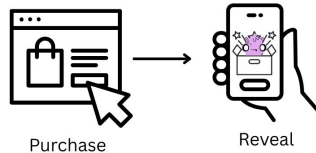
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Traditional Blind Box



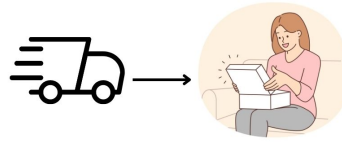
Physical-first suspense

Pop Now



Purchase

Reveal



Shipped

Receive

Digital-first reveal

Our Prototype



Hybrid

Figure 1: We contrast three approaches to blind box unboxing: the traditional blind box (physical-only reveal upon opening), Pop Now (digital reveal prior to physical receipt), and our prototype orchestration families (hybrid narrative experiences that reframe suspense through tempo control, diegetic storytelling, and synchronous social play). Together, these designs demonstrate how unboxing can move beyond gambling metaphors toward richer, more participatory, and socially meaningful experiences.

Abstract

Blind boxes have surged among youth and adults, turning purchase into a social ritual of suspense and reveal. Platforms reshape this ritual with “instant reveal, ship later” flows that separate revelation from possession. Designing these screen-first climaxes is an HCI challenge—balancing pacing, agency, perceived fairness, and co-presence—yet guidance for suspense across digital–physical boundaries is limited. We study Pop Mart’s Pop Now via interviews, co-design, and a lab comparison with synchronized physiological signals. Instant reveals compress arousal into brief peaks and can reduce felt control and embodied meaning, even as they add convenience and certainty. Our hybrid prototype shows how configurable pacing and consequential micro-choices extend the suspense arc and improve perceived control and fairness while preserving efficiency. We contribute the PAC framework, an evaluation pipeline linking physiology, behaviour, and self-report, and three deployable patterns with guardrails for “reveal now, receive later” experiences.



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CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design.**

Keywords

blind box, surprise, digital-physical commerce

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1 Introduction

Blind boxes have surged among youth and adults, turning purchase into a social ritual of suspense and reveal [19]. Rather than simply acquiring a known product, people engage in a designed experience of anticipation, staged revelation, and often public sharing through “unboxing” videos and live streams [16]. As a result, the interface and flow surrounding the reveal—how suspense is paced, what actions are invited, and how the moment is shared—has become part of the product experience itself.

For HCI, blind boxes represent a broader design problem: interactive systems can script emotional trajectories over time, shaping not only what users receive but also how they feel as events unfold. Experience-centric HCI argues that interaction is temporal, affective, and social [20], and that tactile engagement contributes uniquely to meaning-making [14]. The question is not whether suspense is “good” or “bad”, but how to design reveal experiences that balance convenience with ritual depth—supporting agency, legibility, and trust while preserving the joy of surprise.

Platforms increasingly interpose an “instant reveal, ship later” flow that decouples revelation from possession [1, 2]: users learn the outcome immediately online, while the physical item arrives later. This hybrid move relocates the emotional climax online, compresses multisensory buildup, and gives platforms greater control over pacing and the micro-interactions that produce suspense. Yet HCI lacks clear design guidance for orchestrating suspense across digital–physical boundaries—including how pacing, agency, and co-presence should be recomposed when the reveal is no longer embodied. Pop Mart’s Pop Now [3], widely used for *Labubu*, offers a concrete example.

We treat unboxing as a designed ritual embedded in an experiential arc: what matters is not only *what* is revealed, but *how* and *when* it is staged—interaction is temporal, affective, and social [20]; and that tactile engagement contributes uniquely to meaning-making [14]. Pop Now accelerates knowledge and substitutes platform-controlled animation for anticipation. Some collectors value efficiency and certainty, while others experience reduced embodiment and raise fairness and disclosure concerns [4].

This paper examines Pop Now as a case of *hybrid suspense systems* in consumer technology. We ask three questions:

- **RQ1:** How do fans and non-fans perceive and evaluate Pop Now compared with traditional physical unboxing?
- **RQ2:** What design strategies can recompose suspense, agency, and sociality to make hybrid unboxing emotionally rich and ethically sound?
- **RQ3:** How do physiological markers of arousal differ across physical, current digital, and novel hybrid reveal experiences?

We report a three-study, mixed-methods program and synthesize results into a framework and design principles for configurable suspense and emotional continuity in hybrid reveals. We contribute: (1) an empirical account of how fans and non-fans construe agency, fairness, and surprise across reveal formats; (2) three deployable design patterns—*Suspense Loop*, *Narrative Reveal*, and *Social Sync*; and (3) the PAC framework (Pacing, Agency, Co-presence) and an evaluation pipeline coupling physiology, behaviour, and self-report.

Together, these contributions offer HCI guidance to preserve cultural meaning, protect users, and enable new commercial practices.

2 Related Work

2.1 Experience Trajectories, Anticipation, and Surprise

Experience-centred HCI frames interaction as a temporally unfolding, affective narrative [25]. Trajectories describe how interactive events can be orchestrated across time and place to shape

anticipation, climax, and resolution [30]. Studies of surprise in digital interfaces demonstrate that pacing and expectation management strongly influence evaluation [13]. Cognitive psychology’s peak–end heuristic further predicts that users disproportionately remember affective peaks and endings [10], making reveal choreography consequential for memory. Recent empirical work on blind box-like sales mechanisms shows that blind box mechanisms can enhance purchase intention and satisfaction through the mediating effects of curiosity and perceived value [15, 31, 32]. We therefore need to calibrate balances between predictability and surprise. Mechanism and timing matter: the how/when of the reveal—not merely the outcome—shape perceived value.

2.2 Materiality, Ritual, and Social Co-Presence in Unboxing

Physical cues such as texture, weight, and sound contribute to meaning-making in product encounters [9, 24, 26], and packaging functions as a stage for scripted behaviours such as tearing and peeking [27]. Ritual theory [29] and CSCW research on co-presence show how synchrony, shared attention, and responsive performance intensify affect and bonding [6, 23]. Our work leverages these insights in our Social Sync mechanism (synchronous co-reveal) [7]. In parallel, we explore material echoes (packaging elements that align with digital beats on arrival) so that both physical and digital experiences can resonate together. By creating unboxing as a hybrid ritual, these mechanisms knit sensory, temporal, and social elements into a consistent experience.

2.3 Gamification, Gacha/Loot Boxes, and Hybrid Systems

Chance-based monetization research links variable-ratio rewards to arousal and, at extremes, problematic spending [8, 18]. HCI research on dark patterns cautions against obscured odds [21, 22], artificial countdowns [11, 12, 17, 21, 22], and forced scarcity [5, 12, 17, 21, 22]. Hybrid systems where information (reveal) is decoupled from embodiment (possession) require careful cross-channel hand-offs to preserve coherence. In these systems, specifically Pop Mart, the reveal may be an online unboxing of the item, where users shake the box and reveal the character they got. The embodiment comes a few days or weeks later, in the form of a package that users can then physically unbox to reveal the character that they saw on the Pop Now animation. Omnichannel customer-experience research [28] defines this as the synergetic management of touchpoints across phases, with the pre-purchase stage critical for setting expectations and emotional pacing. Our prototype implements exactly such orchestration by staging hints, embedding narrative beats, and aligning post-reveal packaging cues.

3 Method

3.1 Study 1: Interview—How People Compose a Reveal

40 participants spanning prior familiarity with blind box culture (20 fans, 20 non-fans) were recruited. Recruitment targeted variation in age, gender, and collecting experience to surface contrasts between efficiency-seeking and meaning-seeking practices. Semi-structured

interviews asked participants to walk us through a recent reveal or unboxing, describing location, co-presence, timing (lingering or skipping), and perceptions of fairness, excitement, or memorability before, during, and after the instant of knowing. We probed four domains without presupposing technical solutions: pacing, agency, sociality, and trust. Interviews were recorded and transcribed. Two researchers iteratively developed a codebook from an initial open-coding pass, double-coded a subset, reconciled disagreements through discussion, and then applied the stabilized scheme to the remainder. Codes were organized into practice-grounded tensions—speed vs. savouring; pseudo-control vs. tempo authorship; solitary taps vs. co-witnessing; visible odds vs. opaque theatre.

3.2 Study 2: Participatory Design—From Language to Levers

Three mixed-group co-design workshops were conducted with a total of 24 participants (12 fans, 12 non-fans). Each session began with a structured discussion in which participants shared memorable unboxing experiences. The groups collaboratively mapped experiences into discrete “beats” (tease, hint, build, reveal, celebrate, and remember), sketched flows, assembled low-fi prototypes, and critiqued each other’s concepts while thinking aloud. Facilitators kept the scope ethical and analyzable: leave odds untouched; explore pacing, representation, and social scaffolding; make accessibility a default rather than an afterthought. The emphasis was on compositional control, including where to place pauses, what kinds of hints to allow, and how to legitimize witnessing.

3.3 Study 3: Lab Comparison—Conditions, Tasks, and Apparatus

A within-participants design ($N=12$; 6 fans, 6 non-fans) compared three conditions: Physical, Pop Now, and Hybrid. Condition order was counterbalanced with a 3×3 Latin square. Before the first condition, participants completed a neutral practice trial to learn controls and acclimate to sensors. To reduce content confounds, reveal content was matched across conditions. Each condition comprised four phases: a 60-second pre-reveal baseline (quiet sitting, eyes open; device in hand for digital conditions), a reveal phase (Physical: unseal and open upon a “REVEAL” cue; Pop Now: standard instant-reveal flow on a lab device; Hybrid: progress through the Suspense Loop with the Narrative Reveal micro storyline; Social Sync runs in solo mode), 30-seconds post-reveal handling, and ≈ 60 -seconds self-report ratings. A washout interval (90–120 seconds) to allow arousal to return toward baseline; a brief attention check ensured readiness.

Devices and the stimulus laptop were time-synchronized (NTP) at session start. Event markers were dispatched programmatically to the physiology stream and the interaction logger to achieve millisecond alignment. The app logged every state transition, dwell times, and linger/skip actions. In the Physical condition, a table-mounted camera captured hands and package; two coders later annotated reveal onset and opening phases to create stage-aligned dwell times with inter-rater reliability (Cohen’s κ) before full coding.

3.4 Measures and Operationalization

Primary physiological responses were analyzed using EDA and HRV around the reveal onset, capturing peak arousal and sustained engagement. Secondary measures included behavioural logs—stage dwell times, linger/skip actions, total reveal duration, physical coding from video—and post-condition self-reports of perceived control, fairness, excitement, memorability, and overall preference (7-point Likert scale). Reveal onset was coder-annotated for physical stimuli and frame-defined for digital stimuli; baselines were -30 to 0 seconds prior to reveal, with post-reveal windows of 0 – 10 seconds (peak/time-to-peak), and 0 – 30 seconds (arousal dwell), with sensitivity analyses varying window lengths by ± 10 seconds.

3.5 Ethics and Intellectual Property

All studies were approved by our institutional REB. Participation was voluntary with informed consent, and participants could withdraw at any point without penalty.

All figures are author-created schematics that depict interaction timing and flow; no Labubu or Pop Now UI is reproduced. “Labubu” and “Pop Now” are used nominatively for identification only.

4 Results

4.1 Study 1: Interviews—Diverging Construals of Surprise and Agency

People rarely described “the reveal” as a single moment; they described it as a composed journey. Fans collapsed suspense into a practical question—*which variant did I pull?*—using Pop Now as a “clarity machine” to compress decisions and plan trades, while newcomers lingered on eventfulness. Some participants preferred delivery-day revelation even though Pop Now could tell them immediately because “*the waiting becomes part of the story*”. Agency was a key tension. Participants wanted tempo authorship: honest micro-choices to hold, linger, or skip a beat, rather than pseudo-control. Sociality was budding, enacted via synchronized countdowns in Discord and replays, while trust and fairness were critical, with participants requesting visible odds and verifiable logs. These insights shaped the PAC framework: Pacing, Agency, and Co-presence.

4.2 Study 2: Participatory Design—From Language to Mechanics

Participants’ ideas cohered quickly around pacing and anticipation over odds redistribution. Three design families emerged: **Suspense Loop** staged a sequence of cues with tempo authorship; **Narrative Reveal** used a diegetic micro-story to modulate pacing and theme without affecting fairness; **Social Sync** converted solitary taps into shared experiences via virtual private rooms, synchronized countdowns, and post-reveal state celebrations. Digital-physical linkage consisted of two extensions: **Material Echoes**, which mirrored digital beats in the packaging so that opening the parcel completes a narrative begun on screen, and **Arrival-Day Re-enactment**, which allowed the parcel to be scanned to restage the reveal with gentle haptics. Participants endorsed consequential micro-choices over pseudo-control and disclosing odds and allocation rules. These refinements informed the hybrid prototype evaluated in Study 3.

4.3 Study 3: Physiological Comparison—Shaping the Arc of Arousal

The lab study was designed to read the contour of arousal rather than to chase absolute magnitudes. We synchronized EDA and HRV to five event markers, allowing us to see how different reveal designs paced a body through build-up, recognition, and celebration. Physiological measures showed distinct arousal patterns. Physical unboxing produced gradual, sustained peaks; Pop Now spiked sharply at animation resolution, then decayed just as quickly; the hybrid prototype sat in between the two with a longer post-recognition dwell. Area under the curve for EDA: over the first 60 seconds was higher for Physical than Pop Now, and the hybrid prototype matched much of that sustained engagement while remaining briefer; peak amplitude: Pop Now produced the highest instantaneous peak, consistent with a compressed climax; time-to-peak: the shortest for Pop Now, intermediate for Hybrid, and longest for Physical. HRV changes (RMSSD) mirrored these patterns, dipping most for Pop Now in the immediate aftermath of the reveal, compatible with a short, intense burst followed by a quick return toward baseline. Self-reports aligned: Hybrid scored highest on perceived control and fairness, reduced gambling-like impressions, and elicited more save/replay behaviour. Rare pulls raised instantaneous peaks across the board, but did not erase the differences in AUC between conditions.

5 Discussion

Pacing shapes meaning, not intensity. The physiology showed that Pop Now compresses revelation into a micro-event, while physical unboxing sustains a meso-trajectory. The Hybrid reproduced much of the sustained arc without abandoning digital convenience. Interviews explain the mechanism: people want to author tempo. Providing controllable micro-pause through linger taps and narrative beats increased arousal dwell (AUC) and memorability, especially for non-fans. This supports experience-centric HCI's emphasis on temporal orchestration: the reveal is not a point but a sequence of beats.

Unboxing functions as a ritual performance rather than a transaction. While physical cues remain singular in their multisensory richness, our results show that diegetic narrative and celebratory social cues can substitute for part of that anchoring, generating small moments of collective effervescence even through screens. Materiality, therefore, is not binary but distributable across narrative, social, and physical cues.

Agency must be consequential and disclosed. Participants were alert to pseudo-agency: theatrics undermined trust when undisclosed, but welcomed when framed honestly. Replacing pseudo-agency with consequential micro-choices enhanced perceived control and fairness, and also increased arousal dwell and memorability. We distinguish three agency budgets designers can allocate without touching odds: temporal agency, representational agency, and social agency. These budgets should be explicitly disclosed as experiential rather than probabilistic, with odds and allocation rules visible and reviewable.

Hybrid reveals sit at a boundary where gamified anticipation meets material goods, making harm-reduction defaults essential.

Recommended practices include configurable suspense, continuity cues and arrival-day re-enactments, micro-choice agency, and disclosing odds and making reveal logs inspectable. These recommendations are not anti-excitement; they support authored suspense that preserves meaning while reducing gambling-like impressions—an effect strongest among non-fans in the Hybrid condition. Measuring tempo interactions, replay/save rates, and arrival-day scans better capture experiential quality than conversion and sales alone.

5.1 Limitations and Future Work

Our lab study is modest in size and within-participants, limiting power for small effects and generalizability. Social Sync ran in solo mode; therefore, live multi-user contagion was not measured. To standardize the Pop Now baseline for physiology, we isolated the reveal step and preloaded assets to fix animation duration; commerce context was excluded, which may alter anticipation relative to production use. Rarity balancing reduced, but did not eliminate, desirability confounds, and our neutral on-screen/packaging assets (used to avoid IP reproduction) likely underestimate the aesthetic impact of production artwork. EDA is sensitive to motion and temperature, and short-window HRV serves primarily as a triangulation signal. Only adults were recruited; results do not generalize to youth. Future studies will: (i) test live, multi-user synchrony at scale to model amplification/damping; (ii) compare short diegetic wrappers by genre and language; (iii) evaluate packaging-level echoes in field deployments across real shipping intervals; (iv) prototype fairness proofs that are legible to non-technical users; (v) conduct youth-focused studies under a minors-approved protocol; (vi) run longitudinal field experiments to estimate novelty decay, replay behaviour, and retention.

6 Conclusion

Pop Now illustrates a class of hybrid suspense systems where digital revelation is decoupled from physical possession. Through interviews, participatory design, and physiological evaluation, we show that unboxing can be re-composed to balance efficiency with emotional depth. Our hybrid prototype—combining Suspense Loop, Narrative Reveal, and Social Sync—produced arousal patterns closer to physical unboxing while preserving the speed and convenience of digital commerce. We contribute a taxonomy and design principles for configurable suspense and emotional continuity, framing unboxing as a designable ritual across modalities. More broadly, HCI should move beyond simulating physical acts to orchestrate new choreographies of anticipation and celebration that respect diverse user tempos, ethical guardrails, and the social life of things.

References

- [1] [n. d.]. <https://fabfitfun.com/get-the-box>
- [2] [n. d.]. <https://www.footlocker.com/launch101.html>
- [3] [n. d.]. <https://www.popmart.com/ca/activities/329DFD0E>
- [4] 2025. https://www.reddit.com/r/labubu/comments/1ky22c5/i_dont_get_the_appeal_of_pop_now/
- [5] Luiz Adolpho Baroni, Alisson Andrey Puska, Luciana Cardoso de Castro Salgado, and Roberto Pereira. 2021. Dark Patterns: Towards a Socio-technical Approach. In *Proceedings of the XX Brazilian Symposium on Human Factors in Computing Systems (Virtual Event, Brazil) (IHC '21)*. Association for Computing Machinery, New York, NY, USA, Article 15, 7 pages. doi:10.1145/3472301.3484336

- [6] Pernille Bjørn, Morten Esbensen, Rasmus Eskild Jensen, and Stina Matthiesen. 2014. Does Distance Still Matter? Revisiting the CSCW Fundamentals on Distributed Collaboration. *ACM Trans. Comput.-Hum. Interact.* 21, 5, Article 27 (Nov. 2014), 26 pages. doi:10.1145/2670534
- [7] Jin Hyun Cheong, Zainab Molani, Sushmita Sadhukha, and Luke J. Chang. 2023. Synchronized affect in shared experiences strengthens social connection. *Communications Biology* 6, 1 (Oct 2023). doi:10.1038/s42003-023-05461-2
- [8] Pieterjan Declerck. 2025. *Lootboxes and Random Reward Mechanisms*. T.M.C. Asser Press, The Hague, 69–107. doi:10.1007/978-94-6265-691-8_4
- [9] Willemijn S. Elkhuisen, Jeff Love, Stefano Parisi, and Elvin Karana. 2024. On the Role of Materials Experience for Novel Interactions with Digital Representations of Historical Pop-up and Movable Books. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 619, 18 pages. doi:10.1145/3613904.3642142
- [10] BL Fredrickson and D Kahneman. 1993. Duration neglect in retrospective evaluations of affective episodes. *Journal of personality and social psychology* 65, 1 (July 1993), 45–55. doi:10.1037//0022-3514.65.1.45
- [11] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3173574.3174108
- [12] Colin M. Gray, Cristiana Santos, and Nataliia Bielova. 2023. Towards a Preliminary Ontology of Dark Patterns Knowledge. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 286, 9 pages. doi:10.1145/3544549.3585676
- [13] Alice Marlene Gross and Juliane Bürglen. 2014. The Impact of Expectations on User Experience: Surprising the User. In *Proceedings of the 28th International BCS Human Computer Interaction Conference on HCI 2014 - Sand, Sea and Sky - Holiday HCI* (Southport, UK) (BCS-HCI '14). BCS Learning & Development Ltd, Swindon, GBR, 22–31. doi:10.14236/ewic/hci2014.3
- [14] Arminda Guerra. 2025. Craft-Based Methodologies in Human-Computer Interaction: Exploring Interdisciplinary Design Approaches. *Multimodal Technologies and Interaction* 9, 2 (2025). doi:10.3390/mti9020013
- [15] Wen Jinling. 2024. A study on the mechanism of the influence of blind box marketing on consumers' purchase intention. *The EUrASEANs: journal on global socio-economic dynamics* 6(49) (Nov 2024), 131–144. doi:10.35678/2539-5645.6(49).2024.131-144
- [16] Hyosun Kim. 2020. Unpacking unboxing video-viewing motivations: The uses and gratifications perspective and the mediating role of parasocial interaction on purchase intent. *Journal of Interactive Advertising* 20, 3 (2020), 196–208.
- [17] Emre Kocyyigit, Arianna Rossi, and Gabriele Lenzini. 2024. A Systematic Approach for A Reliable Detection of Deceptive Design Patterns Through Measurable HCI Features. In *Proceedings of the 2024 European Symposium on Usable Security (EuroUSEC '24)*. Association for Computing Machinery, New York, NY, USA, 290–308. doi:10.1145/3688459.3688475
- [18] Chanel J. Larche, Katrina Chini, Christopher Lee, Mike J. Dixon, and Myra Fernandes. 2019. Rare loot box rewards trigger larger arousal and reward responses, and greater urge to open more loot boxes. *Journal of Gambling Studies* 37, 1 (Nov 2019), 141–163. doi:10.1007/s10899-019-09913-5
- [19] Cary Lee, Jessica Wyllie, and Stacey Brennan. 2025. Eye-opening! Exploring uncertainty marketing through hedonic blind box collectibles. *Journal of Retailing and Consumer Services* 82 (2025), 104127.
- [20] Theo Mahut, Carole Bouchard, Jean-Francois Omhover, Carole Favart, and Daniel Esquivel. 2017. Interdependency between user experience and interaction: A kansei design approach. *International Journal on Interactive Design and Manufacturing (IJiDeM)* 12, 1 (Feb 2017), 105–132. doi:10.1007/s12008-017-0381-4
- [21] Arunesh Mathur, Gunes Acar, Michael J. Friedman, Eli Lucherini, Jonathan Mayer, Marshini Chetty, and Arvind Narayanan. 2019. Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 81 (Nov. 2019), 32 pages. doi:10.1145/3359183
- [22] Arunesh Mathur, Mihir Kshirsagar, and Jonathan Mayer. 2021. What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 360, 18 pages. doi:10.1145/3411764.3445610
- [23] Reneeta Mogan, Ronald Fischer, and Joseph A. Bulbulia. 2017. To be in synchrony or not? A meta-analysis of synchrony's effects on behavior, perception, cognition and affect. *Journal of Experimental Social Psychology* 72 (2017), 13–20. doi:10.1016/j.jesp.2017.03.009
- [24] Marianna Obrist. 2017. Mastering the Senses in HCI: Towards Multisensory Interfaces. In *Proceedings of the 12th Biannual Conference on Italian SIGCHI Chapter (Cagliari, Italy) (CHIItaly '17)*. Association for Computing Machinery, New York, NY, USA, Article 2, 2 pages. doi:10.1145/3125571.3125603
- [25] Marianna Obrist, Rob Comber, Sriram Subramanian, Betina Piqueras-Fiszman, Carlos Velasco, and Charles Spence. 2014. Temporal, affective, and embodied characteristics of taste experiences: a framework for design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 2853–2862. doi:10.1145/2556288.2557007
- [26] Erica Robles and Mikael Wiberg. 2010. Texturing the "material turn" in interaction design. In *Proceedings of the Fourth International Conference on Tangible, Embedded, and Embodied Interaction* (Cambridge, Massachusetts, USA) (TEI '10). Association for Computing Machinery, New York, NY, USA, 137–144. doi:10.1145/1709886.1709911
- [27] Keith Russell. 2015. Unboxing and the ends of design: A psychology of unpacking. *The International Journal of Designed Objects* 9, 2 (May 2015), 17–23. doi:10.18848/2325-1379/cgip/v09i02/38691
- [28] Si Shi, Yi Wang, Xuanzhu Chen, and Qian Zhang. 2020. Conceptualization of omnichannel customer experience and its impact on shopping intention: A mixed-method approach. *International Journal of Information Management* 50 (2020), 325–336. doi:10.1016/j.ijinfomgt.2019.09.001
- [29] Erika Summers-Effler. 2006. *Ritual Theory*. Springer US, Boston, MA, 135–154. doi:10.1007/978-0-387-30715-2_7
- [30] Raphael Velt, Steve Benford, and Stuart Reeves. 2017. A Survey of the Trajectories Conceptual Framework: Investigating Theory Use in HCI. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 2091–2105. doi:10.1145/3025453.3026022
- [31] Zhiyi Wei and Bin Yu. 2025. Why do you engage in blind box consumption? Exploring the group interactions and psychological motivations in blind box consumption. *Current Psychology* (08 2025), 1–15. doi:10.1007/s12144-025-08150-x
- [32] Yi Zhang and Tianqi Zhang. 2022. The effect of blind box product uncertainty on consumers' purchase intention: The mediating role of perceived value and the moderating role of purchase intention. *Frontiers in Psychology* Volume 13 - 2022 (2022). doi:10.3389/fpsyg.2022.946527